6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

6. A girl standing 6·0 m from a convex mirror sees her virtual image 1·0 m behind the mirror. The focal length of the mirror is : *1*

(A) 6·0 m (B) 1·2 m

(C) 2·4 m (D) 0·8 m

CBSE 26C

(B) 1.2 m

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C 2

20. What is meant by linear magnification of an image formed by a spherical mirror ? Write an expression for the magnification produced by a concave mirror in terms of the object distance and the image distance. CBSE 26C *2*

Linear magnification is the ratio of the size of image to the size of object. Alternatively: $$m = \frac{h_I}{h_O} = \frac{\text{Size of image}}{\text{Size of object}}$$	1
$$m = -\frac{v}{u} = -\frac{\text{Image distance}}{\text{Object distance}}$$	1

25. A thin convex lens of focal length 25 cm is cut into two pieces 0·5 cm parallel and above the principal axis. An object is placed at a distance of 50 cm in front of the smaller part. Find : (i) the position of the image, and (ii) the magnification produced by this part. CBSE 26C 3

25. A thin convex lens of focal length 25 cm is cut into two pieces 0·5 cm parallel and above the principal axis. An object is placed at a distance of 50 cm in front of the smaller part. Find : (i) the position of the image, and (ii) the magnification produced by this part. CBSE 26C 3

(i)	$f = +25\text{ cm},\ u = -50\text{ cm},\ v = ?$	½
	$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$	½
	$\frac{1}{25} = \frac{1}{v} - \frac{1}{-50} = \frac{1}{v} + \frac{1}{50}$	
	$\frac{1}{v} = \frac{1}{25} - \frac{1}{50} = \frac{2-1}{50} = \frac{1}{50}$	½
	$v = +50cm$	½
(ii)	$m = \frac{v}{u}$	½
	$m = \frac{50}{-50} = -1$	½

33. (a) (i) Explain the phenomenon of total internal reflection. State two conditions that must be satisfied for total internal reflection to occur. Derive the relation between the critical angle and the refractive index of a medium.

(ii) A ray of light is travelling from a denser medium of refractive index 1·5 to a rarer medium of refractive index 1·0. Calculate critical angle for total internal reflection. CBSE 26C 5

(i) When a ray of light enters from a denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases at the interface, angle of refraction also increases till the angle of incidence is $\frac{\pi}{2}$ (i.e the refracted ray grazes the surface of the interface). If the angle of incidence is further increased, the incident ray totally gets reflected back. This is called total internal reflection. | 2

Conditions

1. Light must travel from a denser medium to a rarer medium. | ½

2. At the interface, the angle of incidence must be greater than the critical angle. | ½

Relation

When light goes from a denser medium (refractive Index n_2) towards a rarer medium (refractive Index n_2), then for the value of critical angle, we have relation-

$n_2 \sin i_c = n_1 \sin 90^\circ$ | ½

$\sin i_c = \frac{n_1}{n_2}$

or, $\frac{n_2}{n_1} = \frac{1}{\sin i_c}$

or,

$n_{21} = \frac{1}{\sin i_c}$ | ½

(ii) Calculating critical angle

$n_2 = 1.5 = \frac{3}{2}$, $n_1 = 1$

$\sin i_c = \frac{1}{n_{21}} = \frac{2}{3}$ | ½

Therefore,

$i_c = \sin^{-1}\left(\frac{2}{3}\right)$ | ½

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0° (B) 30°

(C) 60° (D) 90°

CBSE 26C

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0° (B) 30°

(C) 60° (D) 90°

CBSE 26C

(A) 0^0

10. A point source in air is placed in front of convex spherical glass surface (n = 1·5 and R = 10 cm). Its real image is formed at the same distance as the surface from the source. The distance of the source from the surface is :

(A) 25 cm

(B) 50 cm

(C) 75 cm

(D) 100 cm

CBSE 26C

10. A point source in air is placed in front of convex spherical glass surface (n = 1·5 and R = 10 cm). Its real image is formed at the same distance as the surface from the source. The distance of the source from the surface is :

(A) 25 cm

(B) 50 cm

(C) 75 cm

(D) 100 cm

CBSE 26C

(B) 50 cm

(b) (i) The lower half of a concave mirror's reflecting surface is covered with an opaque non-reflecting material. What effect will this have on the image of an object placed in front of the mirror ? *1*

(ii) A lens made of glass of refractive index 1·5 disappears, when immersed in a trough of liquid. What is the refractive index of the liquid ?

Give reasons for your answer in each case. CBSE 26C *1*

(b) (i) The lower half of a concave mirror's reflecting surface is covered with an opaque non-reflecting material. What effect will this have on the image of an object placed in front of the mirror ? *1*

(ii) A lens made of glass of refractive index 1·5 disappears, when immersed in a trough of liquid. What is the refractive index of the liquid ?

Give reasons for your answer in each case. CBSE 26C *1*

(i) The image will be formed that of the whole object. However, the intensity of the image will be low (or half)	1
(ii) The refractive index of the liquid = 1.5 Since lens disappears inside the liquid which is possible only when both have same refractive indices.	½ ½

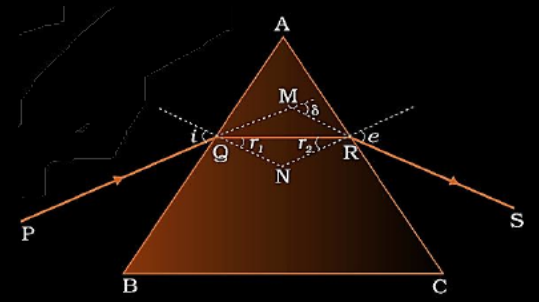
24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. *3*

CBSE 26C

24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. 3

CBSE 26C

In quadrilateral AQNR,



$\angle A + \angle QNR = 180°$

From triangle QNR,

½

$r_1 + r_2 + \angle QNR = 180°$

Comparing the above equations,

$r_1 + r_2 = A$

The total deviation δ is the sum of deviations at the two faces:

$\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - A$ ½

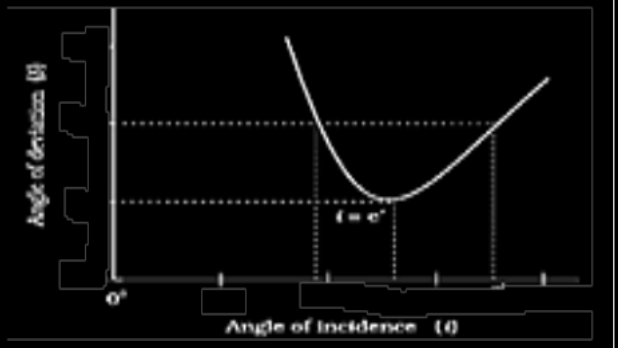
At minimum deviation,

$\delta = D_m$, $i = e$ *which implies* $r_1 = r_2$

$r = \frac{A}{2}$

$D_m = 2i - A$ or $i = \frac{A+D_m}{2}$ ½

The refractive index of the prism is



$n = \frac{\sin i}{\sin r}$ 1

$n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ ½

27. (a) Draw a labelled ray diagram of a refracting telescope in normal adjustment. Briefly explain its working. *2*

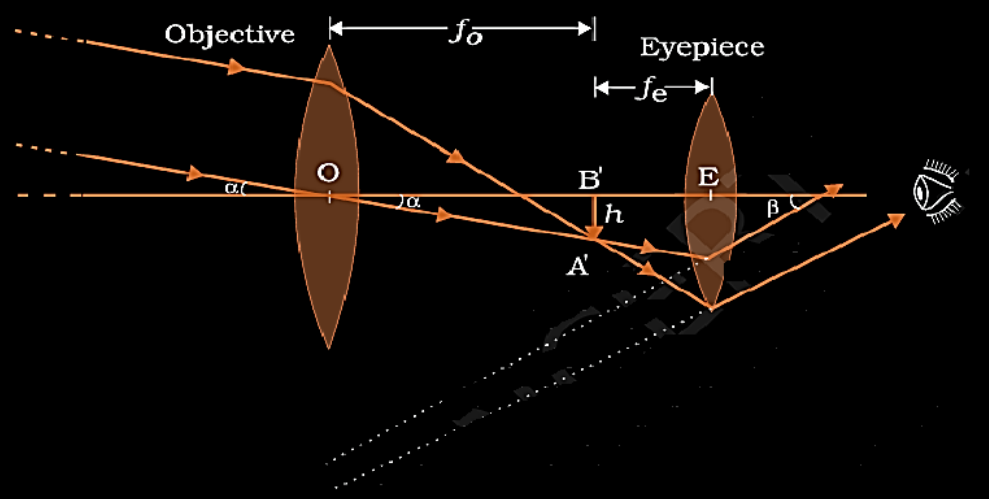
(b) Why is the eyepiece of a telescope of short focal length, while the objective is of large focal length ? Explain. *1*

CBSE 26C

27. (a) Draw a labelled ray diagram of a refracting telescope in normal adjustment. Briefly explain its working. *2*

(b) Why is the eyepiece of a telescope of short focal length, while the objective is of large focal length ? Explain. *1*

CBSE 26C

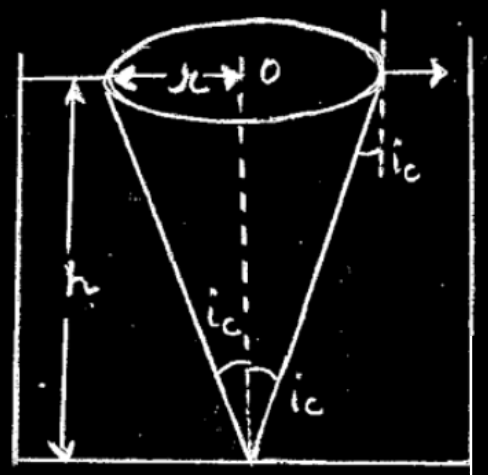


Working: When light from a distant object enters the objective lens, a real image is formed in the tube at its second focal point. The eyepiece magnifies this image, producing a final inverted image.	1
(b) To increase the magnifying power or angular magnification.	1
	1

33. (a) A small bulb is placed at the bottom of a tank containing water to a depth of 1·00 m. Find the area of the surface of water through which light from the bulb can emerge from. The refractive index of water is (4/3). *3*

(b) An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed coaxially at a distance of 10 cm from the lens. If the image formed by this combination coincides with the object itself, draw the ray diagram and find the focal length of the convex mirror. *2*

CBSE 26C



1

$\sin i_c = \frac{1}{n}$ ½

$\tan i_c = \frac{r}{h}$

$r = h \tan i_c$ & $\tan i_c = \frac{1}{\sqrt{n^2-1}}$

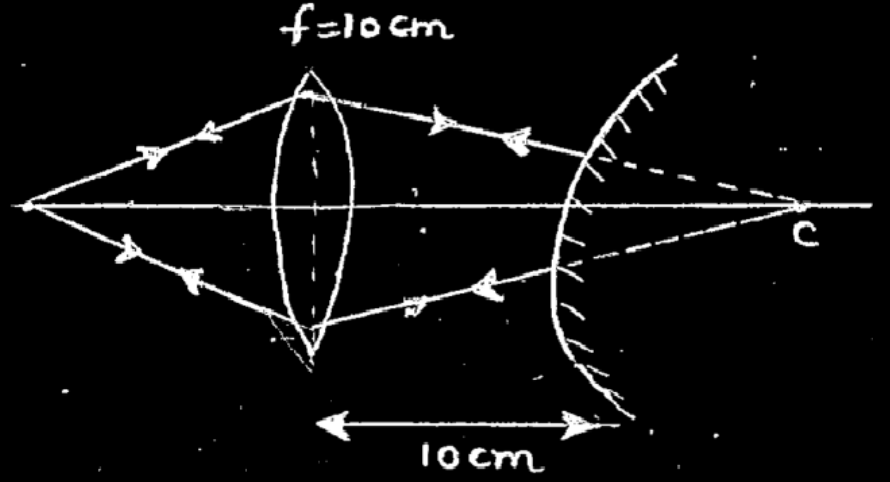
$A = \pi r^2 = \pi h^2 \tan^2 i_c$ ½

$A = \frac{\pi h^2}{n^2-1}$ ½

$A = \frac{\pi \times 1}{\frac{16}{9}-1}$ ½

$= \frac{9\pi}{7}$ m^2 $= \frac{9\times 3.14}{7}$ =4.04 m^2

(b)



1

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{15}$

$= \frac{1}{30}$

$v = 30$ cm ½

Radius of curvature of the mirror,

$R = 30 - 10 = 20$ cm

$f = \frac{R}{2}$ ½

$f = 10$ cm

9. The focal lengths of the objective and the eyepiece of a compound microscope are 2 cm and 3 cm respectively. If they are kept 25 cm apart, the tube length of the microscope is :

(A) 23 cm (B) 22 cm

(C) 20 cm (D) 30 cm

CBSE 26C

9. The focal lengths of the objective and the eyepiece of a compound microscope are 2 cm and 3 cm respectively. If they are kept 25 cm apart, the tube length of the microscope is :

(A) 23 cm (B) 22 cm

(C) 20 cm (D) 30 cm

CBSE 26C

(C) 20 cm

24. (a) The distance of an object from a concave mirror is increased from 12 cm to 15 cm. It is observed that the new image becomes $\left(\frac{2}{5}\right)^{\text{th}}$ the size of the initial image. Calculate the focal length of the mirror.

(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C *3*

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(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C *3*

(a)

For $u = -12\ cm$

$$m_1 = \frac{h_1}{h_o} = \frac{f}{f+12} \quad \ldots(1)$$ ½

For $u = -15\ cm$

$$m_2 = \frac{h_2}{h_o} = \frac{f}{f+15} \quad \ldots(2)$$ ½

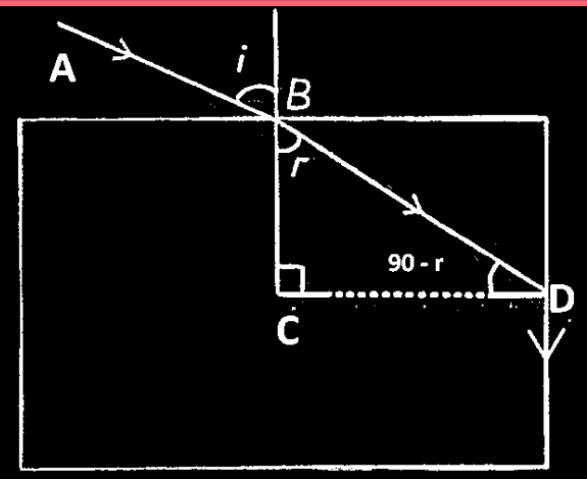
Dividing (1) by (2)

$$\frac{m_1}{m_2} = \frac{h_1}{h_2} = \frac{f+15}{f+12}$$

$$h_2 = h_1\left(\frac{2}{3}\right)$$

On solving,

$$f = -10\ cm$$ ½



At air-glass interface,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin r}$$

$$\sin r = \frac{\sin 45^\circ}{n} = \frac{1}{\sqrt{2}\,n} \quad \ldots(1)$$ ½

At glass-air interface, light suffers Total Internal Reflection (TIR) ($i_C = 90°\text{-}r$).

$$n = \frac{1}{\sin i_C}$$

$$n = \frac{1}{\sin(90^\circ - r)} = \frac{1}{\cos r}$$

$$n = \frac{1}{cosr}$$

$$\cos r = \frac{1}{n} \quad \ldots(2)$$ ½

Squaring and adding (1) and (2)

$$1 = \frac{1}{2n^2} + \frac{1}{n^2}$$

On solving,

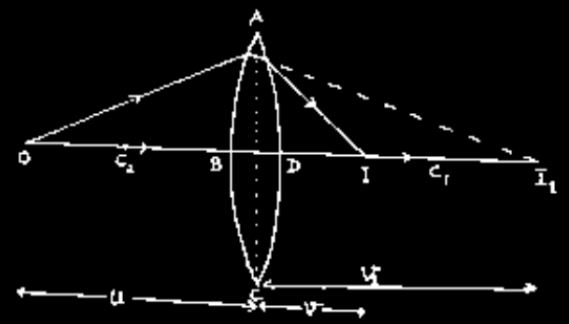
$$n = \sqrt{\frac{3}{2}}$$ ½

27. A thin convex lens with its two surfaces of radii of curvature R_1 and R_2 is made of a material of refractive μ_2. A medium of refractive index μ_1 ($< \mu_2$) surrounds it. Using a ray diagram, derive the Lens Maker's formula when a point object placed on the principal axis in front of the surface of radius of curvature R_1 produces an image I on the other side of the lens. *3*

CBSE 26C

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CBSE 26C



1

The first refracting surface forms the image of O at I_1.

For refraction through the first refracting surface (ABC),

$\frac{\mu_2}{v_1} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1}$ .....(1) ½

Similarly for refraction through second refracting surface ADC, the image I_1 acts as a virtual object and forms the final image at I.

$\frac{\mu_1}{v} - \frac{\mu_2}{v_1} = \frac{\mu_1 - \mu_2}{R_2}$ .....(2) ½

Adding equations (1) and (2),

$\frac{\mu_1}{v} - \frac{\mu_1}{u} = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

Suppose the object is at infinity i.e. $u = \infty$, and image at f.

$\frac{\mu_1}{f} = (\mu_2 - \mu_1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$

$\frac{1}{f} = \left(\frac{\mu_2}{\mu_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$ ½

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

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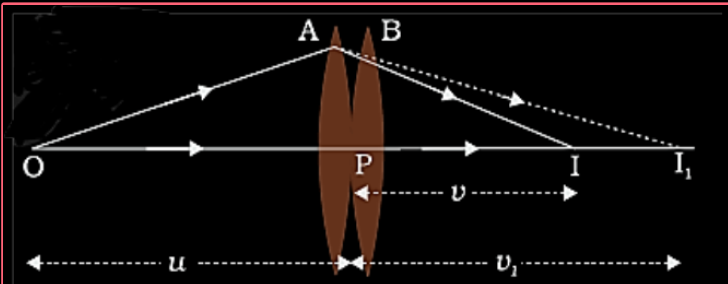
(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

(B) π/4

24. Two thin converging lenses of focal lengths f_1 and f_2 are placed coaxially in contact. Draw a ray diagram to show the image formation by the combination. Hence, derive the expression for the focal length of the combination. CBSE 26C 3

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1

For the image formed by the first lens,

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad \text{---------- (1)}$$ ½

For the second lens,

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad \text{---------- (2)}$$ ½

Adding equations (1) and (2),

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_1} + \frac{1}{f_2} \quad \text{---------- (3)}$$ ½

For the two-lens system being equivalent to a single lens of focal length f, the equation (3) becomes

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{------ (4)}$$

On comparing equations (3) and (4),

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$ ½

27. Two converging lenses L_1 and L_2, of focal lengths 144 cm and 6 cm respectively, are arranged coaxially in a tube 147 cm apart. The arrangement is used to view a far off tree, keeping the eye close to L_2.

(i) Draw the ray diagram depicting image formation by the combination.

(ii) How far from L_2 will the final image be formed ?

(iii) Calculate the magnification produced by the combination. *3*

CBSE 26C

(ii) $L = 147\text{ cm} > f_1 + f_2 = (150\text{ cm})$

The first image is formed between the focus and lens L_2.

i.e. $u_2 < f_2$

$L = f_1 + |u_2|$

$|u_2| = L - f_1 = 147 - 144 = 3\text{ cm}$ ½

For L_2

$u_2 = -3\text{ cm},\ f_2 = 6\text{ cm}$

$v_2 = \frac{u_2 f_2}{u_2 + f_2}$

$v_2 = \frac{(-3)\times 6}{-3+6}$ ½

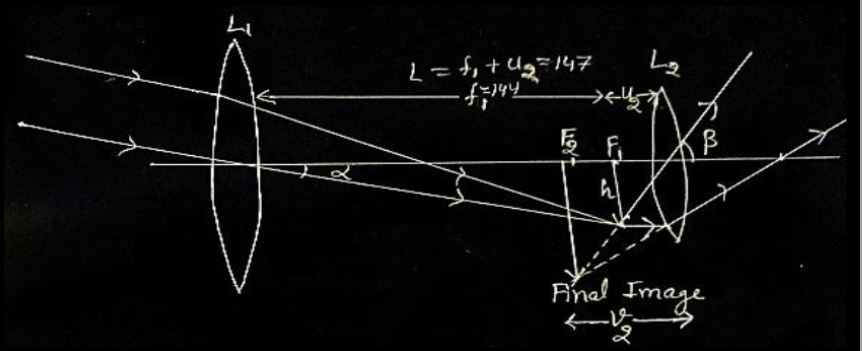
$= -6\text{ cm}$

Final image is formed at 6 cm from L_2 at left side.

(iii) Magnification, $m = \frac{tan\beta}{tan\alpha}$

$= \frac{h/u_2}{h/f_1}$ ½

(i)



1

$m = \frac{f_1}{u_2}$

$m = f_1\left(\frac{1}{v_2} - \frac{1}{f_2}\right)$

$m = 144\left(\frac{1}{-6} - \frac{1}{6}\right)$

$m = -48$ ½

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

$$n = \frac{\text{speed of light in vacuum(c)}}{\text{speed of light in medium(v)}} \quad \ldots\ldots\ (i)$$ ½

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$ ½

$$n = \frac{12.5}{9} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

From (i) and (ii)

$$v = \frac{3 \times 10^8 \times 9}{12.5}$$ ½

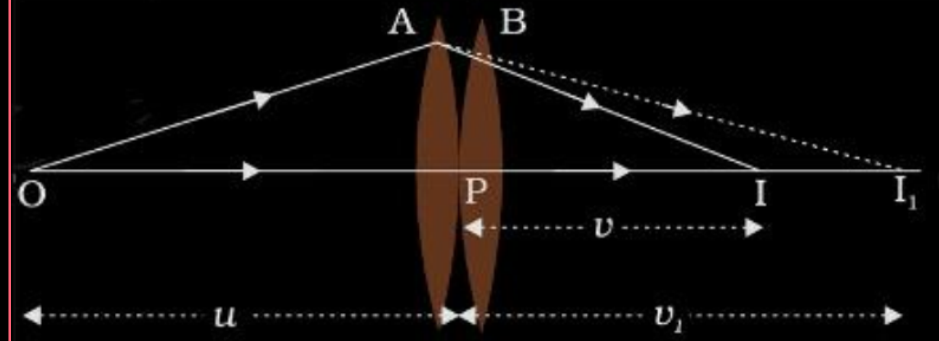
$$= 2.16 \times 10^8 \text{ m/s}$$ ½

20. Two thin lenses of focal length f_1 and f_2 are placed in contact with each other coaxially. Prove that the focal length f of the combination is given by $f = \dfrac{f_1 f_2}{f_1 + f_2}$. 2

CBSE 2026

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CBSE 2026



½

For the image formed by lens A

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1} \quad \text{.....(i)}$$

½

For the image formed by second lens B

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2} \quad \text{.....(ii)}$$

½

Adding (i)and(ii)

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f_2} + \frac{1}{f_1}$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

$$f = \frac{f_1 f_2}{f_1 + f_2}$$

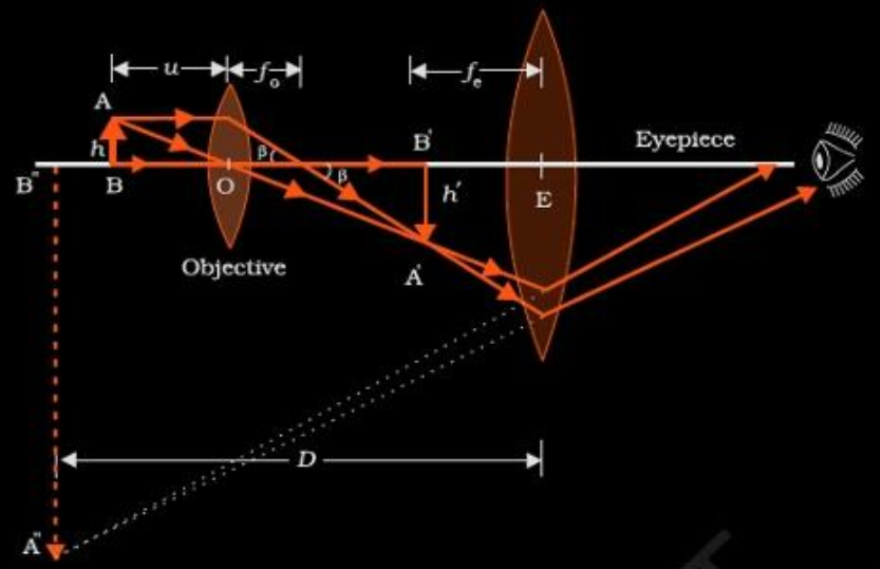
½

(b) (i) With the help of a ray diagram, describe the construction and working of a compound microscope.

(ii) (I) The real image of an object placed between f and 2f from a convex lens can be seen on a screen placed at the image location. If the screen is removed, is the image still there ? Explain.

(II) Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances ? Explain. CBSE 2026 5

(i)



1

Construction: It consists of two convex lenses, objective lens with small aperture placed near the object and the eye piece with large aperture placed near the eye. Both lenses have small focal length. These lenses are placed coaxially in a narrow tube with rack and pinion arrangement.

1

Working:
The lens near to the object called objective forms a real, inverted & magnified image of the object. This serves as the object for eye piece which produces the final image, enlarged & virtual.

1

(ii) I Yes ½

Real image is formed by actual intersection of rays. ½

II Yes ½

If the object is virtual. ½

Note: 1. If students do not write construction award full marks for well-labelled diagram representing objective lens, eye piece, along with image position at near point or normal adjustment.

2. Deduct ½ mark if arrows are not drawn.

21. A point source, in air, is placed at a distance of 6 cm in front of a convex spherical surface (n = 1·5 and radius of curvature = 24 cm). Find the position and nature of the image formed. CBSE 2026 *2*

21. A point source, in air, is placed at a distance of 6 cm in front of a convex spherical surface (n = 1·5 and radius of curvature = 24 cm). Find the position and nature of the image formed. CBSE 2026 *2*

$$\frac{n_2 - n_1}{R} = \frac{n_2}{v} - \frac{n_1}{u} \quad \text{½}$$

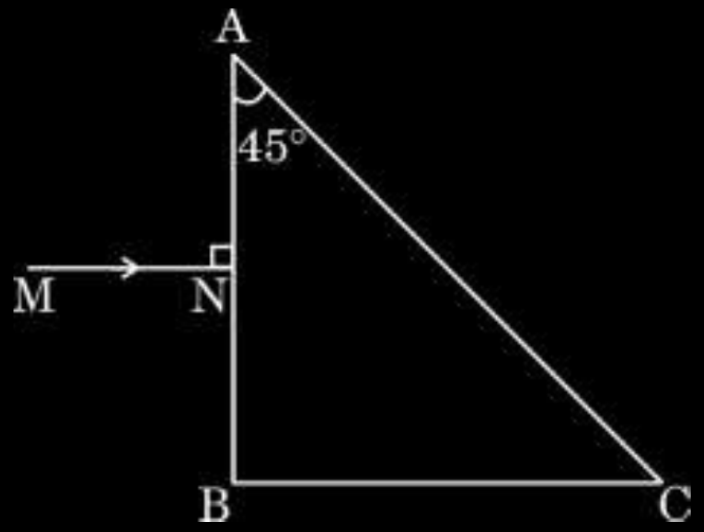
$$\frac{1.5 - 1.0}{24} = \frac{1.5}{v} - \frac{1.0}{-6} \quad \text{½}$$

After solving

$$v = \frac{-72}{7} \approx -10.28cm \quad \text{½}$$

Nature: Virtual ½

18. A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$. *2*



CBSE 2026

(i)

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$$

$$i_c = 45^0$$

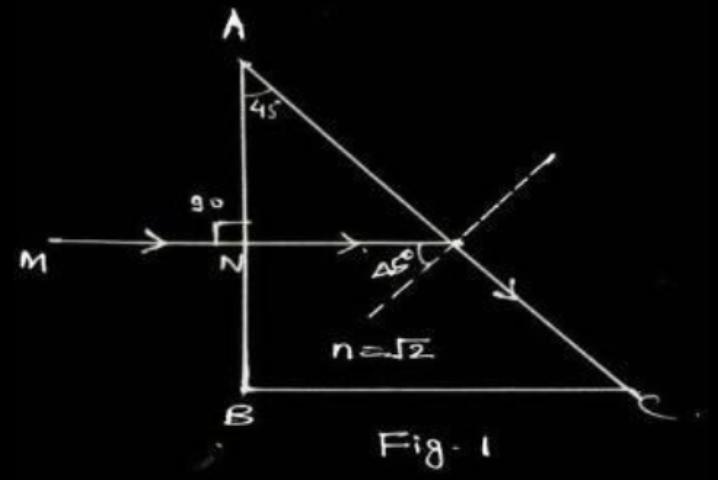


Fig. 1

1

Since $i = i_c$ so the emergent ray grazes alongside AC.

(ii)

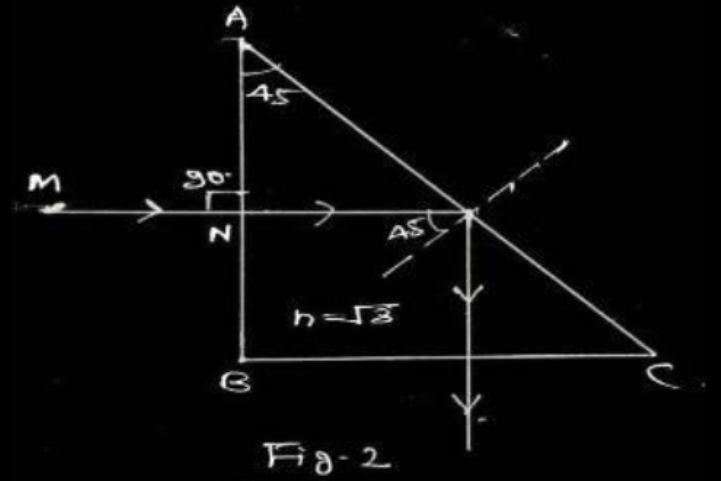


Fig. 2

1

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$$

$$i_c < 45^0$$

Since $i > i_c$ so, the ray suffers from Total Internal Reflection (TIR)

15. *Assertion (A) :* When a convex lens made of glass is immersed in water, its converging power increases.

Reason (R) : The focal length of a lens depends only on the radii of curvature of its two faces.

CBSE 2026

15. *Assertion (A) :* When a convex lens made of glass is immersed in water, its converging power increases.

Reason (R) : The focal length of a lens depends only on the radii of curvature of its two faces.

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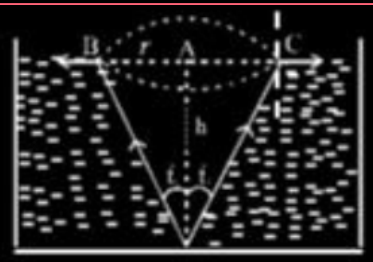
(D) Both assertion(A) and Reason (R) are false.

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

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19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

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$r = h \times \tan i \quad [\text{as } i = i_c]$ ½

$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}} \Rightarrow i_c = 45^\circ$ ½

$r = h$

$\text{Area} = \pi r^2$ ½

$= 3.14 \times (1)^2$

$\text{Area} = 3.14\ m^2$ ½

30. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

(i) The images formed by the objective lens and the eyepiece are respectively : *1*

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

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30. An astronomical telescope consists of two converging lenses. One of them of large aperture and large focal length is called objective lens and the other one, of smaller focal length and smaller aperture is called the eyepiece. It is used to see distant objects which are not seen clearly with naked eyes. The image formed by the objective lens acts as an object for the eyepiece and the final image produced by the eyepiece is magnified.

(i) The images formed by the objective lens and the eyepiece are respectively : *1*

(A) virtual, real

(B) real, virtual

(C) virtual, virtual

(D) real, real

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(i) (B) real, virtual

(ii) The magnification produced by the telescope does ***not*** depend upon the : *1*

(A) colour of light

(B) focal length of objective lens

(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

(iii) Which of the following statements is ***not*** correct for this telescope ? *1*

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) The distance between two lenses is more than ($f_o + f_e$).

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

CBSE 2026

(ii) The magnification produced by the telescope does ***not*** depend upon the : *1*

(A) colour of light

(B) focal length of objective lens

(C) focal length of eyepiece

(D) apertures of objective lens and eyepiece

(iii) Which of the following statements is ***not*** correct for this telescope ? *1*

(A) The focal length of objective lens (f_o) is larger than the focal length of eyepiece (f_e).

(B) Its magnifying power can be increased by increasing the focal length of objective lens (f_o).

(C) The distance between two lenses is more than ($f_o + f_e$).

(D) The magnifying power can be decreased by increasing the focal length of eyepiece.

(ii) (D) apertures of objective lens and eyepiece

(iii) (C) The distance between two lenses is more than $(f_0 + f_e)$

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(iv) (a) An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of : *1*

(A) 84 cm (B) 76 cm

(C) 20 cm (D) 320 cm

OR

(b) Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be : *1*

(A) 320 (B) 84

(C) 76 (D) 20

(iv) (a) An astronomical telescope has objective lens and eyepiece of focal lengths 80 cm and 4 cm respectively. To view the image in normal adjustment, the lenses must be separated by a distance of : *1*

(A) 84 cm (B) 76 cm

(C) 20 cm (D) 320 cm

OR

(b) Consider the telescope described in question (iv) (a). Its magnifying power in normal adjustment will be : *1*

(A) 320 (B) 84

(C) 76 (D) 20

(iv) (A) 84 cm

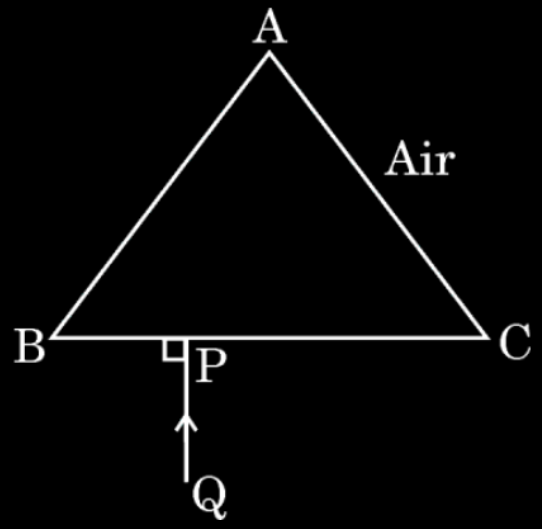
OR

(D) 20

31. (a) (i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum derivation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation. *5*



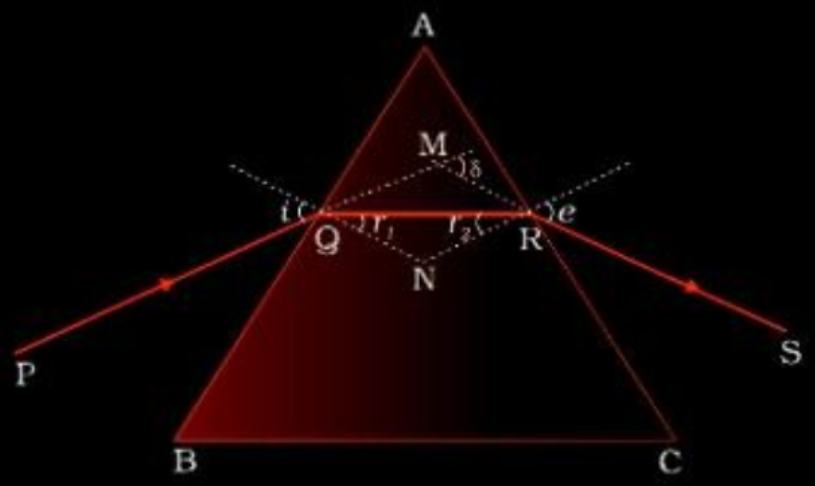
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(i) Refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium. — 1

Alternatively

$$n_{21} = \frac{c}{v}$$

(ii)



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180°.

$\angle A + \angle QNR = 180°$

From the triangle QNR,

$r_1 + r_2 + \angle QNR = 180°$

Comparing these two equations, we get

$r_1 + r_2 = A$ .....(i) — ½

The total deviation δ is the sum of the deviations at the two faces.

$\delta = i + e - A$ .....(ii) — ½

For $\delta = \delta_m$; i = e which implies $r_1 = r_2$

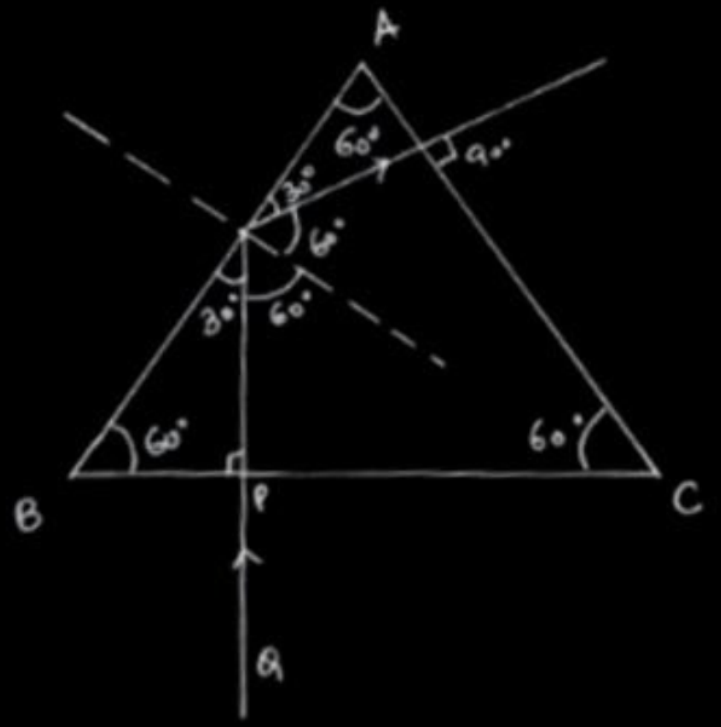
From equation (i) and (ii) — ½

$$r = \frac{A}{2}; i = \frac{A + \delta_m}{2}$$

Refractive index of the prism

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\frac{(A + \delta_m)}{2}}{\sin\frac{A}{2}}$$ — ½

(iii)

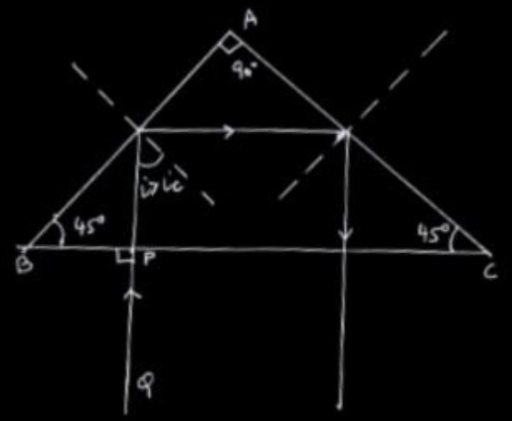


1

The ray QP strikes the face BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. It strikes the face AC normally and comes out undeviated.

1

Alternatively



The ray QP strikes the surface BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. As at face AC, $i > i_c$, it undergoes TIR. The ray comes out of the face BC undeviated.

(Award full credit of this part if the student takes the value of A other than 60° or 90° and traces the path correctly and according to the angle taken)

OR

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

CBSE 2026

19. A concave mirror of radius of curvature 16 cm produces two-times enlarged real image of an object kept in front of it. Find the position of the object. *2*

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Given: $R = -16$ cm (concave mirror)

$$m = -\frac{v}{u} = -2$$

$$v = 2u$$ ½

Now for concave mirror,

$f = -8\text{cm}, v = 2u, u = ?$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$ ½

$$\frac{1}{2u} + \frac{1}{u} = \frac{1}{-8}$$ ½

$$\frac{1+2}{2u} = \frac{1}{-8}$$

$$\frac{2u}{3} = -8$$

$$u = -\frac{24}{2} = -12\text{cm}$$

$$\therefore u = -12\text{cm}$$ ½

20. A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25. *2*

CBSE 2026

20. A convex lens of refractive index 1·5 has a focal length of 20 cm in air. Find its nature and focal length when it is immersed in a transparent liquid of refractive index 1·25. *2*

CBSE 2026

$$\frac{1}{f} = ({}^{a}\mu_{g} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \qquad ½$$

$$\frac{1}{f'} = ({}^{\ell}\mu_{g} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

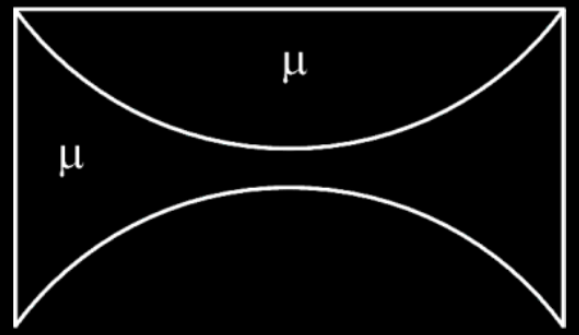
$$\frac{f'}{f} = \frac{({}^{a}\mu_{g} - 1)}{({}^{\ell}\mu_{g} - 1)} = \frac{1.5 - 1}{\left(\frac{1.5}{1.25} - 1\right)} = \frac{0.5}{0.25} \times 1.25 \qquad ½$$

$$\frac{f'}{f} = 2.5$$

$$\text{or } f' = 2.5 \times 20 = 50\text{cm} \qquad ½$$

Nature ⟶ converging. ½

4. A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is :



(A) $\frac{R}{\mu - 1}$

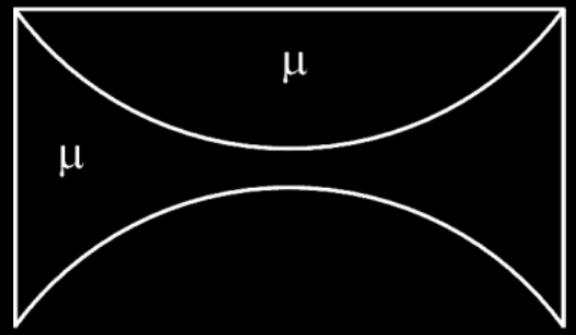
(B) $-\frac{R}{\mu - 1}$

(C) $\frac{2R}{\mu - 1}$

(D) $-\frac{2R}{\mu - 1}$

CBSE 2026

4. A thin plano-convex lens and a thin equi-concave lens are kept coaxially in contact as shown in the figure. Assuming both the lenses are made of glass of refractive index μ, and R is the radius of curvature of each curved surface, the focal length of the combination is :



(A) $\frac{R}{\mu - 1}$

(B) $-\frac{R}{\mu - 1}$

(C) $\frac{2R}{\mu - 1}$

(D) $-\frac{2R}{\mu - 1}$

CBSE 2026

(B) $\frac{-R}{\mu-1}$

6. A telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6·0 cm. The magnifying power and the length of telescope tube will be respectively :

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

CBSE 2026

6. A telescope has an objective lens of focal length 144 cm and an eyepiece of focal length 6·0 cm. The magnifying power and the length of telescope tube will be respectively :

(A) 24, 150 cm

(B) 42, 138 cm

(C) 24, 138 cm

(D) 42, 150 cm

CBSE 2026

(A) 24, 150 cm

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

(A) Twinkling of stars

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

CBSE 2026

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

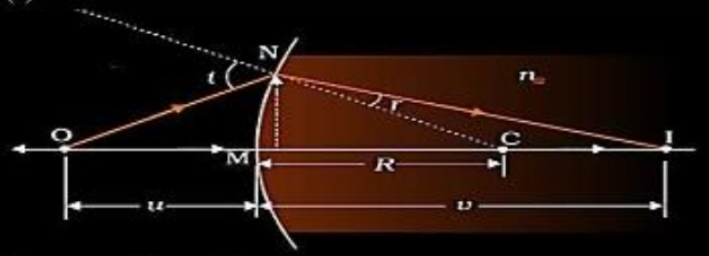
CBSE 2026

$\sin i_C = \frac{1}{n}$	
$= \frac{2}{3}$	1
For glass – water surface	
$\frac{3}{2}\sin i_C = \frac{4}{3}\sin r$	½
$\frac{3}{2} \times \frac{2}{3} = \frac{4}{3}\sin r$	
$\sin r = \frac{3}{4}$	
For water – air surface.	½
$\frac{4}{3}\sin r = \sin e$	
$\frac{4}{3} \times \frac{3}{4} = \sin e$	½
$\sin e = 1$	
$e = 90^\circ$	½

33. (a) (i) A point object is kept in front of a convex spherical surface of radius of curvature R. Draw the ray diagram to show the formation of image and derive the relation between the object and image distance (u and v) in terms of refractive index n of the medium and R.

(ii) A convex lens of focal length of 20 cm is used to form the image of an object placed 30 cm away from the lens. Find the position and nature of the image formed. CBSE 2026 *5*

(i)



1

For small angles

$\tan\angle NOM = \frac{MN}{MO}$

$\tan\angle NCM = \frac{MN}{MC}$

$\tan\angle NIM = \frac{MN}{MI}$

$i = \angle NOM + \angle NCM$

$= \frac{MN}{MO} + \frac{MN}{OC}$ ----------(1) ½

Similarly

$r = \angle NCM - \angle NIM$

$= \frac{MN}{MC} - \frac{MN}{MI}$ ----------(2) ½

By Snell's law for small angles

$l \times i = n \times r$

substituting i and r from (1) and (2)

$\frac{1}{MO} + \frac{n}{MI} = \frac{n-1}{MC}$ ----------(3) ½

Using cartesian sign conventions

MO = -u, MI = +v, MC = +R

Substituting in (3)

$\frac{n}{v} - \frac{1}{u} = \frac{n-1}{R}$ ½

(Note : Give full credit of this part if student derives the relation by forming virtual image.)

(ii) $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ ½

$\frac{1}{20} = \frac{1}{v} - \frac{1}{-30}$ ½

v = 60 cm ½

Nature of image: Real and inverted ½

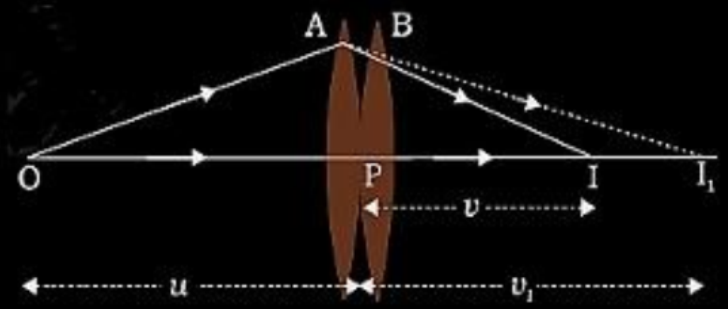
(b) (i) Two thin converging lenses of focal length f_1 and f_2 are placed coaxially in contact. Derive expression for the focal length of the combination.

CBSE 2026

(b) (i) Two thin converging lenses of focal length f_1 and f_2 are placed coaxially in contact. Derive expression for the focal length of the combination.

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(i)



½

For image formed by first lens A

$$\frac{1}{v_1} - \frac{1}{u} = \frac{1}{f_1}$$

½

For image formed by second lens B

$$\frac{1}{v} - \frac{1}{v_1} = \frac{1}{f_2}$$

½

Adding and comparing with lens formula

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

½

3. An equiconvex lens of focal length 15 cm is cut into two equal plano-convex halves. The focal length of each half will be :

(A) 30 cm (B) 20 cm

(C) 15 cm (D) 7·5 cm

CBSE 2026

3. An equiconvex lens of focal length 15 cm is cut into two equal plano-convex halves. The focal length of each half will be :

(A) 30 cm (B) 20 cm

(C) 15 cm (D) 7·5 cm

CBSE 2026

(A) 30 cm

5. A plano-convex lens, made of glass (n = 1·5) has a curved surface whose radius is 40 cm. The image formed of an object is of the same size as that of the object. The distance of the object from the lens is :

(A) 40 cm (B) 80 cm

(C) 160 cm (D) 200 cm

CBSE 2026

5. A plano-convex lens, made of glass (n = 1·5) has a curved surface whose radius is 40 cm. The image formed of an object is of the same size as that of the object. The distance of the object from the lens is :

(A) 40 cm (B) 80 cm

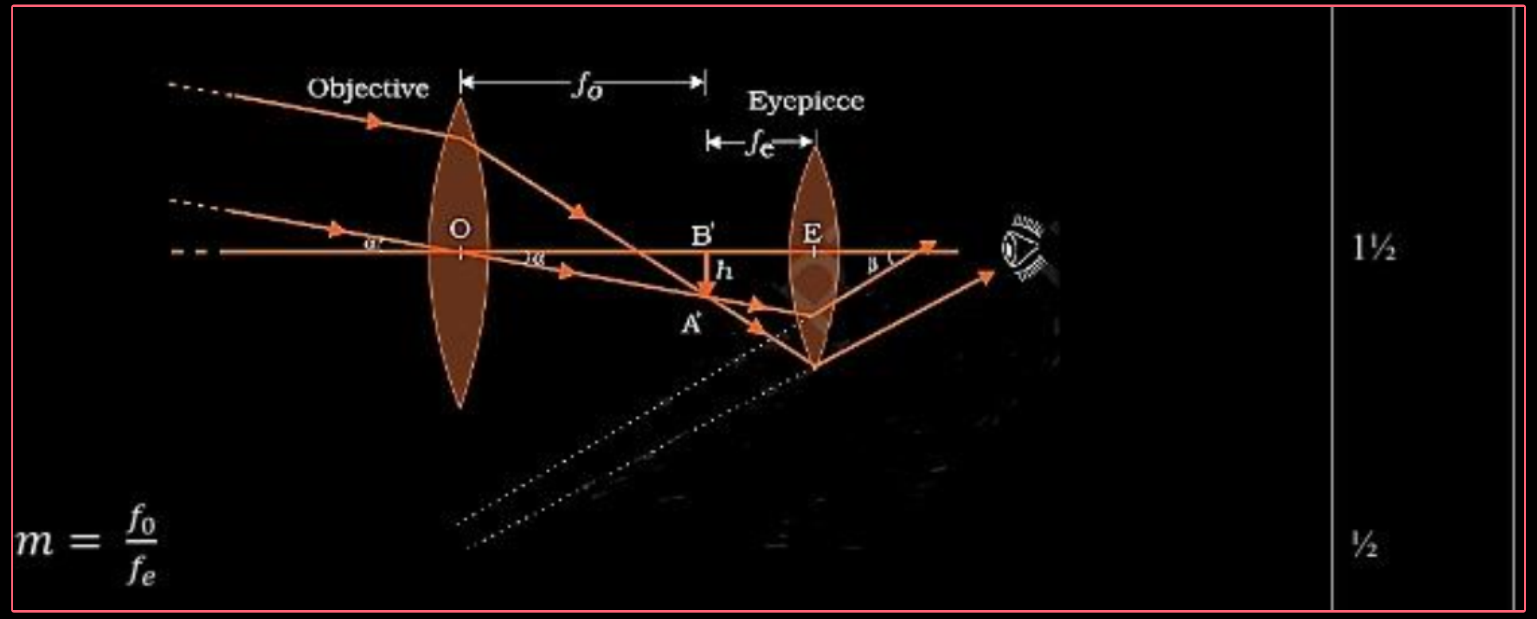
(C) 160 cm (D) 200 cm

CBSE 2026

(C) 160 cm

20. Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power. CBSE 2026 *2*

20. Draw a labelled ray diagram of an astronomical telescope when final image is formed at infinity. Write the expression for its magnifying power. CBSE 2026 *2*

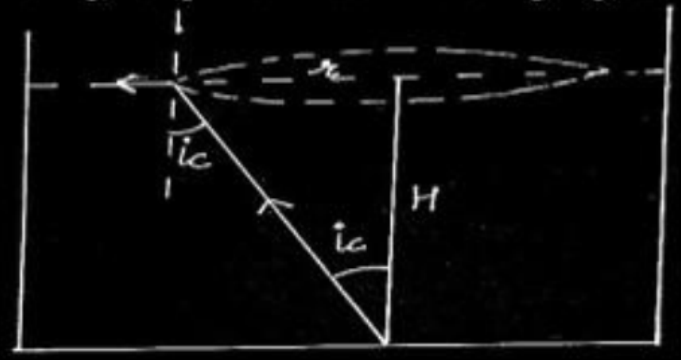


1½

½

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

Critical angle: The angle of incidence corresponding to the angle of refraction 90° for a given pair of media when light goes from denser to rarer medium.	1
	½
$\text{Sin } i_c = \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + H^2}}$	½
$\mu^2 r^2 = r^2 + H^2$	½
$(\mu^2 - 1)r^2 = H^2$	
$r = \frac{H}{\sqrt{\mu^2 - 1}}$	½

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

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9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

(C) Total internal reflection occurs for both green and blue lights.

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

(a) As ray of light passes symmetrically through the prism

$\angle r_1 = \angle r_2 = r$, $i = e$ ½

$\delta_m = 2i - A$ ½

$= 2 \times 45° - 60°$

$= 30°$ ½

(b) $r = \frac{A}{2} = 30°$

$\mu = \sin \frac{(\frac{A + \delta_m}{2})}{sin\frac{A}{2}}$ ½

$= \sin \frac{(\frac{60° + 30°}{2})}{sin30°}$ ½

$= \frac{sin45°}{sin30°}$

$= \sqrt{2}$ ½

(Alternatively)

$\mu_a \sin i = \mu_g \sin r$

$1 \times \sin 45° = \mu_g \times \sin 30°$

$\mu_g = \sqrt{2}$

4. The process that allows light to travel through a tube which is not straight is – 1

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

4. The process that allows light to travel through a tube which is not straight is – 1

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

(D) total internal reflection of light

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

CBSE 2026

18. (a) A ray of light travelling along the principal axis is incident on a spherical mirror and reflected back along the same path. Explain, why ? **2**

OR

(b) Write all the sign convention followed for a convex lens. **2**

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When a ray of light travelling along the principal axis is incident on a spherical mirror, then incident light is normal to the surface of spherical mirror, so it reflects back along the same path. **2**

Alternatively,

The angle of incidence along the principal axis is zero therefore from the laws of reflection, the angle of reflection will be zero. Thus, it reflects back along the same path.

1. All the distances are measured from the optical center of the lens.
2. Distances measured along the direction of the incident ray are taken as positive.
3. Distance measured opposite to the direction of incident ray are taken as negative.
4. Height above the principal axis is taken as positive and height below the principal axis is taken as negative.

½ ×4

19. A small object is placed in air in front of a convex glass surface (n = 1.5 and radius of curvature 20 cm) at a distance of 50 cm from it. Find the position and nature of the image. CBSE 2026 **2**

19. A small object is placed in air in front of a convex glass surface (n = 1.5 and radius of curvature 20 cm) at a distance of 50 cm from it. Find the position and nature of the image. CBSE 2026 **2**

$\frac{-n_1}{u}+\frac{n_2}{v}=\frac{n_2-n_1}{R}$	½
$\frac{-1}{(-50)}+\frac{1.5}{v}=\frac{1.5-1}{20}$	½
$\frac{1}{50}+\frac{1.5}{v}=\frac{0.5}{20}$	
$\frac{1.5}{v}=\frac{1}{40}-\frac{1}{50}$	
v = 300 cm or 3m	½
Nature of the image is real and inverted	½

(a) Briefly explain the working of a reflecting type telescope. Write its two advantages over a refracting telescope.

(b) A telescope consists of two lenses of focal length 125 cm and 5 cm. Find its magnifying power when final image is at (i) infinity (ii) at a distance of 25 cm from eye. CBSE 2026 **5**

(i) Parallel rays from a distant object (star) enter in the telescopes parallel to principal axis of the mirror tend to collect at the focus of the concave mirror. These reflected rays encounter a secondary convex mirror before meeting at focus, the convex mirror reflected them to the eye piece. Final image is seen through the eye piece. Final image formed is inverted w.r.t the object. **2**

Advantages:
(1) Free from chromatic aberration
(2) Free from Spherical aberration
(3) Require Less mechanical support **½ + ½**
(4) low Cost
(Any Two)

(ii) Magnifying power,

(I) $m = \frac{-f_o}{f_e} = \frac{-125}{5} = -25$ **½ + ½**

½

(II) $m = -\frac{f_o}{f_e}\left(1+\frac{f_e}{D}\right)$

$m = \frac{-125}{5}\left(1+\frac{5}{25}\right) = -30$ **½**

NOTE:- Award full marks if student does not use negative sign in the formula of magnification.

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

Angle of incidence $i = \frac{3}{4}A = 45^o$	1/2
At minimum deviation $r = \frac{A}{2} = 30^o$	1/2
$n = \frac{\sin i}{\sin r} = \frac{\sin 45^o}{\sin 30^o}$	
$n = \sqrt{2}$	1/2
$n = \frac{c}{v}$	
$v = \frac{3}{\sqrt{2}} \times 10^8$ m/s = 2.12×10^8m/s	1/2

21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

21. A 5 cm long pencil is placed along the principal axis of a concave mirror of focal length 20 cm such that its nearest end is at a distance of 25 cm from the mirror. Calculate the length of the image of the pencil. CBSE 2026 **2**

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	$^1/_2$
$\frac{1}{v_1} = \frac{1}{f} - \frac{1}{u_1}$	
$= \frac{1}{(-20)} - \frac{1}{(-25)}$ (For near end of the pencil)	
$v_1 = -100\text{cm}$	$^1/_2$
$\frac{1}{v_2} = \frac{1}{(-20)} - \frac{1}{(-30)}$ (For far end of the pencil)	
$v_2 = -60$ cm	$^1/_2$
Length of the image L = $\lvert v_1 \rvert - \lvert v_2 \rvert = 40\text{cm}$	$^1/_2$

30. In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two positions of the lens. The distance between these two positions of the lens is d.

(i) The value of d is **1**

(A) $\sqrt{D(D-4f)}$ (B) $\sqrt{D(D-2f)}$

(C) $2\sqrt{Df}$ (D) $\sqrt{D(D-f)}$

(ii) Compared to the size of the object, the images formed in the two positions of the lens are respectively **1**

(A) reduced, enlarged (B) reduced, reduced

(C) enlarged, enlarged (D) enlarged, reduced

CBSE 2026

30. In an experiment with convex lens of focal length f, the screen is fixed at a distance D from the object. A student slowly moves the lens away from the object towards the screen and finds that she is able to form sharp image of the object for two positions of the lens. The distance between these two positions of the lens is d.

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(ii) Compared to the size of the object, the images formed in the two positions of the lens are respectively **1**

(A) reduced, enlarged (B) reduced, reduced

(C) enlarged, enlarged (D) enlarged, reduced

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(i) (A) $\sqrt{D(D-4f)}$

(ii) (D) Enlarged, reduced

(iii) If the distance between object and screen is 80.00 cm and the lens forms sharp images at two positions separated by 20.00 cm., the focal length of convex lens is **1**

(A) 15.50 cm (B) 18.75 cm

(C) 20.50 cm (D) 22.75 cm

(iv) Consider a convex lens of focal length 15 cms. For which of the following values of object-screen distance, two positions of the object can be found to obtain sharp image on the screen ? **1**

(A) 45 cm (B) 50 cm

(C) 55 cm (D) 65 cm

OR

(iv) A thin convex lens of focal length 10 cm and another thin lens of focal length 'f' are placed coaxially in contact. If the power of their combination is $\frac{10}{3}$ D, the value of 'f' is **1**

(A) –15 cm (B) –10 cm

(C) –20 cm (D) –30 cm

(iii) If the distance between object and screen is 80.00 cm and the lens forms sharp images at two positions separated by 20.00 cm., the focal length of convex lens is **1**

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(A) –15 cm (B) –10 cm

(C) –20 cm (D) –30 cm

(iii) (B) 18.75 cm

(iv) (D) 65cm

OR

(A) -15 cm

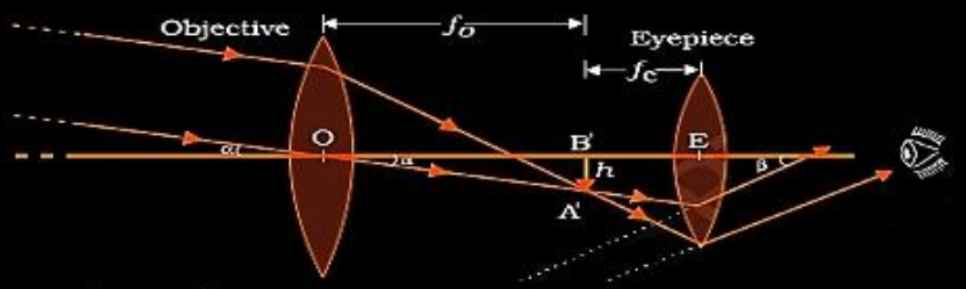
(a) Draw a labelled ray diagram of a refracting telescope when it forms image of a distant object at infinity. Derive expression for its magnifying power.

(b) (i) In a telescope the objective has much larger aperture than the eye piece. Why ?

(ii) Write two advantages of reflecting telescope over refracting telescope.

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(a)



1

- $m = \frac{\beta}{\alpha} \approx \frac{\tan\beta}{\tan\alpha}$ 1

$= \frac{h/f_e}{h/f_o}$ $^1/_2$

$= \frac{f_o}{f_e}$ $^1/_2$

(b) (i) With larger aperture even the fainter objects can be observed clearly. 1

Alternatively:

With larger aperture, the light gathering power and its resolution / resolving power increases.

Alternatively:

Large aperture helps in correcting more light from distant object which forms a sharper/ brighter images.

(ii)
- There is no chromatic aberration in a mirror.
- Spherical aberration in case of parabolic reflectors is removed.
- Requires less mechanical support
- Cost effective
- Brighter / sharper image is formed

(Any two from above or any other appropriate advantages) ($^1/_2$+$^1/_2$))

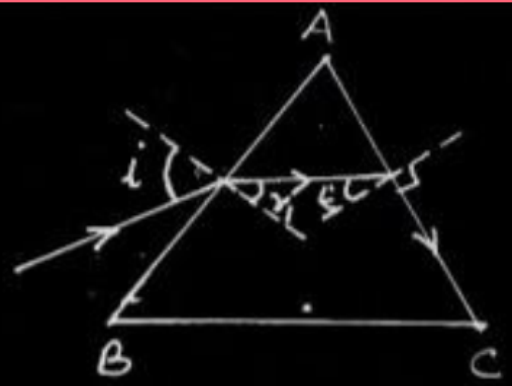
19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

$\sqrt{2}(\sin i_c) = 1$ — ½

$(\sin i_c) = \frac{1}{\sqrt{2}}$

$i_c = 45^0$



$r_1 + r_2 = A$ (Here $r_2 = i_c$) — ½

$r_1 = A - i_c$

$= 60^0 - 45^0$

$= 15^0$ — ½

$\text{Sin } i = \sqrt{2}\sin 15^0$

$i = \sin^{-1}[\sqrt{2}\sin 15^0] = \sin^{-1}(0.366)$ — 1/2

21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

CBSE 2026

21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

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1

From Snell's Law

$n_1 . \sin i = n_2 . sinr$ ½

$A = r_1 + r_2$

$A = r_1 \quad \because r_2 = 0$

$1 \times sin2A = n \times sinA$

$2\ sinA\ cosA = n \times sinA$

$n = 2cosA$ ½

9. A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be **1**

(A) 20 cm (B) 30 cm

(C) 40 cm (D) 5 cm

CBSE 2026

9. A concave lens of focal length 10 cm is cut into two identical plano-concave lenses. The focal length of each lens will be **1**

(A) 20 cm (B) 30 cm

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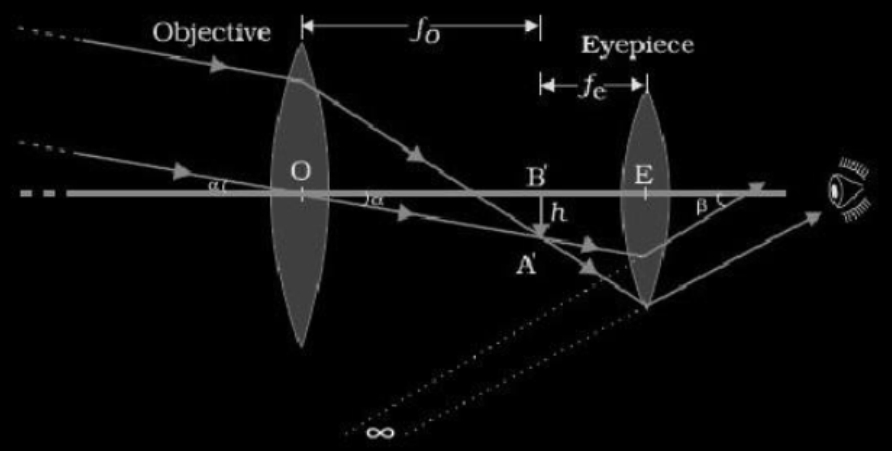
CBSE 2026

(A) 20 cm

27. (a) Draw the ray diagram to show the image formation by a refracting telescope and write the expression for angular magnification for the telescope in normal adjustment.

(b) Give two reasons to explain why a reflecting telescope is preferred over a refracting telescope. CBSE 2026 **3**

(a)



1½

$$|m| = \frac{f_0}{f_e}$$

½

(b)

Reasons:

(i) No Chromatic aberration

(ii) Less mechanical support is required to install it.

½ + ½

(iii) Image formed is brighter and clear.

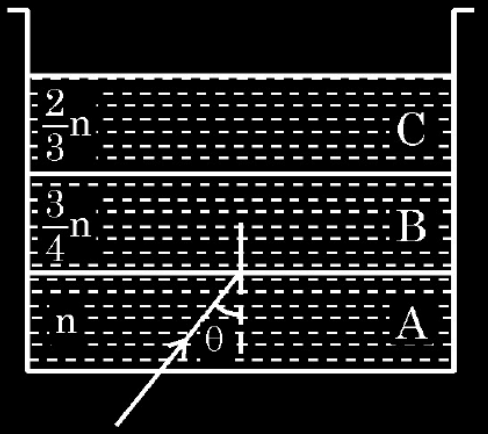
(iv) Cost efficient.

(v) No / less Spherical aberration

[Any two reasons]

28. (a) State the two conditions under which total internal reflection occurs. **3**

(b)



A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n, $\frac{3}{4}$ n and $\frac{2}{3}$ n, respectively. A laser beam is incident at the interface between A and B at an angle θ as shown in figure. Prove that the beam does not enter region C at all for $\sin\theta \geq \frac{2}{3}$.

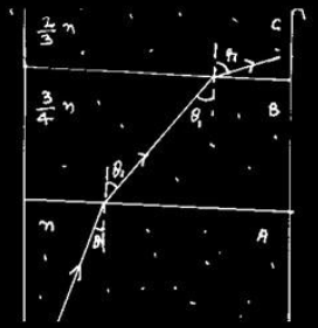
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(a) (i) Light must travel from denser medium to rarer medium. ½

(ii) Angle of incidence must be greater than Critical angle. ½

$i > i_c$

(b)



For refraction at AB interface

$$\frac{\sin\theta}{\sin\theta_1} = \frac{3n/4}{n}$$

$$\sin\theta = \frac{3}{4}\sin\theta_1 \quad \text{---------- (1)}$$ ½

For refraction at BC interface

$$\frac{\sin\theta_1}{\sin r} = \frac{2n/3}{3n/4}$$

$$\sin\theta_1 = \frac{8}{9}\sin r \quad \text{---------- (2)}$$ ½

From equations (1) and (2)

$$\frac{3}{2}\sin\theta = \sin r$$ ½

for $\sin\theta = \frac{2}{3}$

$\sin r = 1$ i.e $r = 90^0$ ½

Ray grazes the surface

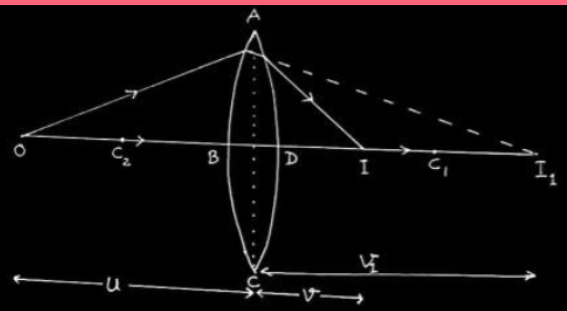
Hence for $\sin\theta \geq \frac{2}{3}$ does not enter the region C at all

32. (a) Using the relation for refraction at a curved spherical surface, derive the expression for lens maker's formula. **5**

(b) Three lenses L_1, L_2 and L_3, each of focal length 40 cm, are placed coaxially. The distance between L_1 and L_2 and between L_2 and L_3 are 120 cm and 20 cm respectively. An object is kept at a distance of 80 cm to the left of lens L_1.

Find the distance of the final image formed from the object. CBSE 2026

(a)



1

The first refracting surface forms the image of the object O at I_1

For refraction from first interface ABC

$$\frac{n_1}{OB}+\frac{n_2}{BI_1}=\frac{n_2-n_1}{BC_1} \qquad \text{---------- (1)}$$ ½

Similarly for refraction from second interface ADC. The image I_1 acts as a virtual object for the second surface ADC

$$-\frac{n_2}{DI_1}+\frac{n_1}{DI}=\frac{n_2-n_1}{DC_2} \qquad \text{---------- (2)}$$ ½

For thin lens $BI_1 = DI_1$

By adding equation (1) and (2)

$$\frac{n_1}{OB}+\frac{n_1}{DI}=(n_2-n_1)\left[\frac{1}{BC_1}+\frac{1}{DC_2}\right]$$

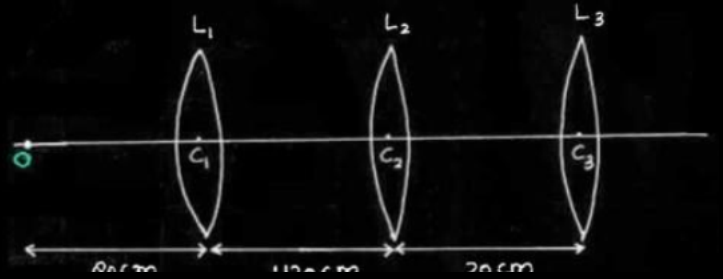
$$-\frac{n_1}{u}+\frac{n_1}{v}=(n_2-n_1)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$ ½

$$\frac{1}{v}-\frac{1}{u}=\left(\frac{n_2}{n_1}-1\right)\left[\frac{1}{R_1}-\frac{1}{R_2}\right]$$

If object is kept at ∞, image will form at focus hence

$$\frac{1}{f}=\left(\frac{n_2}{n_1}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$$ ½

(b)



For lens L_1

$$\frac{1}{f}=\frac{1}{v}-\frac{1}{u}$$

$$\frac{1}{40}=\frac{1}{v_1}+\frac{1}{80}$$

$$\frac{1}{40}=\frac{1}{v_1}+\frac{1}{80}$$

$$\frac{1}{v_1}=\frac{1}{80}$$

$v_1 = 80\ \text{cm}$

For lens L_2

$u_2 = 120-80$

$u_2 = 40\ \text{cm}$

$$\frac{1}{40}=\frac{1}{v_2}+\frac{1}{40}$$

$$\frac{1}{v_2}=0$$

$v_2 = \infty$

For lens L_3

$u_3 = \infty$

$$\frac{1}{40}=\frac{1}{v_3}+\frac{1}{\infty}$$

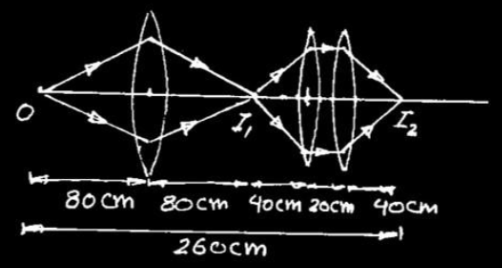
$v_3 = 40\ \text{cm}$

Distance between final image and object

= 80+120+20+40

=260 cm

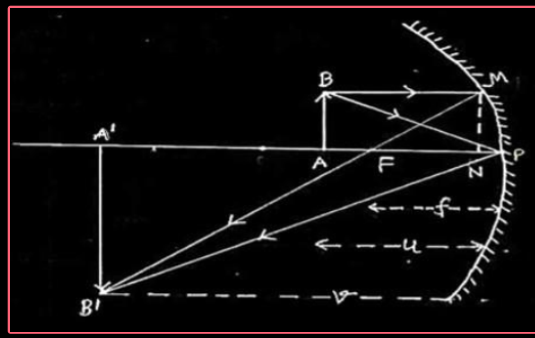
Alternatively:



(a) Draw a ray diagram to show the image formation by a concave mirror when the object is kept between its focus and the centre of curvature. Using this diagram, derive the mirror formula.

(b) A concave mirror produces a two times magnified virtual image of an object kept 10 cm in front of it. Calculate the focal length of the mirror.

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1

ΔBAP and $\Delta B'A'P$ are similar then

$$\frac{BA}{B'A'} = \frac{AP}{A'P} \qquad \text{----------(1)}$$ ½

$\because BA = MN$

ΔMNF and $\Delta B'A'F$ are similar then

$$\frac{MN}{B'A'} = \frac{NF}{A'F}$$

$$\frac{BA}{B'A'} = \frac{NF}{A'F} = \frac{PF}{A'F}$$ (N is very close to P and BA = MN) --------(2) ½

From equation (1) and (2)

$$\frac{AP}{A'P} = \frac{PF}{A'F}$$

$$\frac{-u}{-v} = \frac{-f}{-v+f}$$ ½

$$-uv + uf = -vf$$

$$\frac{-uv}{uvf} + \frac{uf}{uvf} = \frac{-vf}{uvf}$$

$$\frac{-1}{f} + \frac{1}{v} = \frac{1}{-u}$$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$ ½

Note: Award full marks if student derives mirror formula by using any other justified method

(b) $m = \frac{-v}{u}$	½
$m = 2$	
$\because v = -2u \Rightarrow v = -2(-10) = 20\,cm$	½
$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$	
$\frac{1}{20} - \frac{1}{10} = \frac{1}{f}$	½
$\frac{1-2}{20} = \frac{1}{f}$	
$f = -20\,cm$	½

5. A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is **1**

(A) zero

(B) $\frac{20}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

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5. A concave lens of focal length 40 cm is coaxially in contact with two convex lenses, each of focal length 20 cm, on each side. The focal length of the combination is **1**

(A) zero

(B) $\frac{20}{3}$ cm

(C) $-\frac{20}{3}$ cm

(D) $\frac{40}{3}$ cm

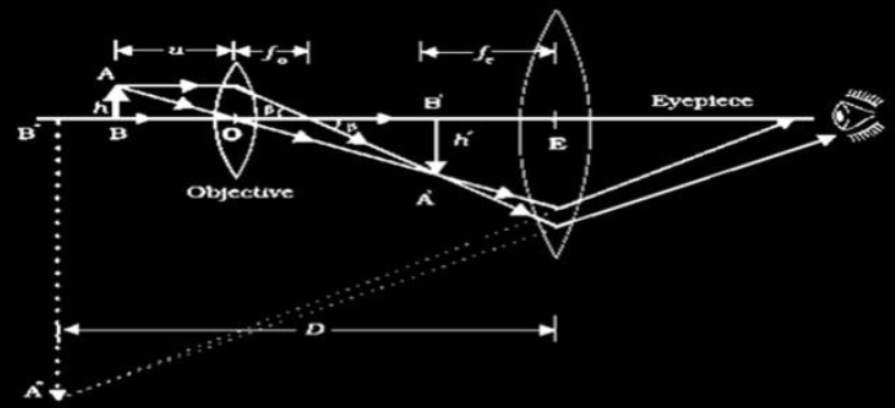
CBSE 2026

(D) $\frac{40}{3}$ cm

28. Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case. CBSE 2026 **3**

28. Draw a labelled ray diagram showing the formation of image by a compound microscope when final image is formed at least distance of distinct vision. Derive an expression for its magnifying power for this case.

CBSE 2026 **3**



1½

Magnification due to the objective lens $m_0 = \frac{h'}{h} = \frac{L}{f_0}$ ½

using $\tan\beta = \left(\frac{h}{f_0}\right) = \left(\frac{h'}{L}\right)$ where L is the length of the tube

When final image is formed at least distance of distinct vision ,then magnification produced by eye lens $m_e = \left(1 + \frac{D}{f_e}\right)$ ½

Total magnification $m = m_o \times m_e$ ½

$$m = \frac{L}{f_0}\left(1 + \frac{D}{fe}\right)$$

Note: Give full marks for any other alternative method.

7. Light from a small object in air falls on a spherical glass surface (n = 1.5) of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as : **1**

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

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7. Light from a small object in air falls on a spherical glass surface (n = 1.5) of radius of curvature R. A real image of the object will be formed if the object distance u is related to R as : **1**

(A) $u < \frac{R}{2}$

(B) $\frac{R}{2} < u < R$

(C) $R < u < 2R$

(D) $u > 2R$

CBSE 2026

(D) u>2R

28. (a) Why does one prefer to view the image formed at infinity than that formed at near point in microscope/telescope ? **3**

(b) Consider lenses L_1, L_2 and L_3 as specified in the following table. Which of them will you select as objective and eyepiece for constructing best possible (i) telescope (ii) compound microscope ? Give reason for your answer.

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Lens	L_1	L_2	L_3
Power	6 D	3 D	10 D
Aperture	1 cm	8 cm	1 cm

(a) In the case of normal adjustment, the eye muscles remain relaxed while they are strained in the case when image is formed at near point while viewing the object, hence it is preferred.	½
(b) • For Telescope	
Lense L_2 is suitable as objective Lens	½
Lense L_3 is suitable as eyepiece	½
• For Compound microscope	
Lense L_3 is suitable as objective lens	½
Lense L_1 is suitable as eyepiece	½
Reason • For telescope, objective lens should have large aperture and large focal length	
• For microscope, objective and eye lens should have moderate aperture but the objective should have large power.	½